Mahatma Gandhi University, Kottayam
Fifth Semester – MG5DSEFSQ305: Processing Technology of Fruits and Vegetables
Regular Examination – October 2026

Answer key

Part A – Multiple-Choice Questions

Q. No.	Answer
1	(b) Horizontally
2	(a) Ice-crystal damage to cell walls
3	(b) Moisture
4	(a) Fruit pulp, sugar and pectin
5	(a) Exocarp
6	(b) Heat processing and hermetic sealing
7	(b) Facilitate skin removal
8	(a) Beetroot
9	(b) Treatment with electrolyzed water
10	(a) Beneficial microorganisms
11	(b) Microbial reduction and surface sanitation
12	(b) Beer and wine
13	(b) Blanching
14	(a) Evaporation of water
15	(d) All of these
16	(a) Moisture loss
17	(b) Sugar
18	(d) Sublimation
19	(d) Polyphenol oxidase
20	(c) Dehydration

Part B – Short Answer Questions

21. Write a note on factors affecting jelly preparation.

Jelly preparation depends on the quality of fruit juice, pectin concentration, sugar content, acidity and cooking conditions. Fruits should contain sufficient pectin and suitable natural acids for proper gel formation. Sugar contributes to sweetness and helps establish the conditions required for gel formation. The pectin concentration influences the firmness of the jelly, while acidity affects the strength and formation of the gel network. Cooking should be controlled to achieve the correct concentration without excessive heating, which may damage pectin or affect flavour

and colour. Proper straining, filling and sealing are also important for producing a clear, firm and stable jelly.

22. List the major pigments present in fruits and vegetables and state their characteristic colours.

The major pigments in fruits and vegetables include chlorophylls, carotenoids, anthocyanins, betalains and flavonoids. Chlorophylls produce green colours and are found in leafy vegetables such as spinach. Carotenoids produce yellow, orange and red colours; examples include beta-carotene in carrots and lycopene in tomatoes. Anthocyanins produce red, purple and blue colours in fruits such as grapes, berries and plums. Betalains provide red-violet and yellow colours and are particularly characteristic of beetroot. Flavonoids, including certain flavones and flavonols, contribute pale yellow colours and other aspects of plant appearance. These pigments influence the colour, attractiveness and perceived quality of fruits and vegetables.

23. Explain simultaneous heat transfer and mass transfer during drying.

Drying involves simultaneous heat transfer and mass transfer. Heat is supplied to the food by hot air, a heated surface or another energy source. This heat raises the temperature of the material and provides the energy needed to evaporate moisture. At the same time, water moves from the interior of the food towards the surface, where it evaporates into the surrounding air. The water vapour is then carried away by the drying medium. The rate of drying depends on temperature, air velocity, relative humidity, surface area, food thickness and the initial moisture content. Proper control of these factors helps achieve efficient drying while limiting undesirable changes in colour, flavour, texture and nutritional quality.

24. Explain fermented beverages and classify them with suitable examples.

Fermented beverages are drinks produced when microorganisms convert sugars or other fermentable substances into alcohol, organic acids and other products. Alcoholic fermented beverages include wine, produced mainly by the fermentation of grape or other fruit sugars, and cider, produced from fermented apple juice. Beer is produced by fermenting a prepared cereal-based wort, usually made from malted grains. Non-alcoholic or low-alcohol fermented beverages include products such as kombucha and certain traditional cereal-based beverages, depending on their preparation and final alcohol content. Fermentation develops characteristic flavours and aromas and may improve product keeping quality. The type of raw material, microorganisms, fermentation conditions and processing method determine the characteristics of the final beverage.

25. Explain the different methods used in cleaning fruits and vegetables.

Cleaning fruits and vegetables removes soil, dust, plant debris, insects and surface contaminants before further processing. Dry cleaning methods include brushing, wiping, screening and air cleaning, which are suitable for removing loose dirt from appropriate produce. Wet cleaning involves washing with potable water through immersion, spraying or water-flume systems. Mechanical agitation may improve the removal of adhering dirt, provided it does not damage delicate produce. Sanitizing treatments, such as approved chlorine-based solutions or other

permitted sanitizers, may be used under controlled conditions to reduce surface microorganisms. Electrolyzed water and ozone may also be used in suitable processing systems. The method selected depends on the type of produce, contamination level and sensitivity to physical damage. Proper drainage and hygienic handling after washing are necessary to prevent recontamination.

26. Describe the physiological, flavour, taste, colour, texture and nutritional changes occurring during refrigerated storage of fruits and vegetables.

During refrigerated storage, low temperatures generally slow respiration, transpiration, ripening and microbial growth, thereby extending the storage life of fruits and vegetables. However, chilling-sensitive produce may develop chilling injury when stored below its suitable temperature. Flavour and aroma may change because the production and release of volatile compounds are altered during storage. Taste may be affected by changes in sugars, organic acids and other soluble compounds. Colour changes associated with ripening and pigment degradation generally occur more slowly, although some deterioration may continue. Texture may change due to moisture loss, softening and modifications in cell-wall components. Nutrients such as vitamin C may gradually decrease during storage, depending on temperature, duration and handling conditions. Maintaining suitable temperature and humidity helps preserve freshness and quality.

27. Explain the processing of wine with a flow chart.

Wine is produced by fermenting the sugars present in grape juice or other suitable fruit juice. First, ripe and healthy fruits are selected, sorted and washed when appropriate. The fruits are crushed to release the juice, and the must is prepared according to the required wine type. Suitable yeast is added to initiate fermentation, during which sugars are converted mainly into ethanol and carbon dioxide. After fermentation, the wine is separated from the sediment by racking and may undergo clarification, maturation and stabilization. Finally, the wine is filtered when required, bottled and sealed.

Flow chart:

Selection of ripe fruits → Sorting and cleaning → Crushing and juice extraction → Must preparation and adjustment → Yeast addition → Fermentation → Racking → Clarification and maturation → Stabilization and filtration → Bottling and sealing.

28. What are food preservatives? Explain their importance in fruit processing.

Food preservatives are substances or treatments used to delay spoilage and maintain the safety and quality of foods during storage. In fruit processing, preservatives help control the growth of spoilage microorganisms and extend the shelf life of products such as juices, squashes, jams and other fruit preparations. Permitted preservatives, including potassium metabisulphite and sodium benzoate, may be used in suitable products according to applicable regulations and prescribed limits. Other preservation approaches include sugar concentration, acidity control, heat treatment, refrigeration and drying. Preservatives may help maintain flavour, colour and product

stability, but their effectiveness depends on the food's pH, composition, processing conditions and packaging. Only approved preservatives should be used at legally permitted levels.

Part C – Essay Questions

29. Discuss the principles and applications of controlled atmosphere packaging and modified atmosphere packaging in fresh produce storage.

Modified atmosphere packaging (MAP) and controlled atmosphere storage (CA) are techniques used to extend the storage life of fresh fruits and vegetables by modifying the gaseous environment around the produce. In MAP, the atmosphere inside a package is modified through gas flushing, suitable packaging materials and the interaction between produce respiration and gas transmission through the film. Oxygen is generally reduced and carbon dioxide is adjusted to levels suitable for the particular commodity. This slows respiration and may delay ripening, senescence and microbial deterioration. The composition of the atmosphere can change during storage.

In controlled atmosphere storage, oxygen, carbon dioxide, temperature and humidity are monitored and adjusted to maintain specified conditions throughout storage. This method is commonly used for the long-term storage of suitable fruits, particularly apples. Both methods require commodity-specific conditions because excessively low oxygen or excessively high carbon dioxide can cause off-flavours, fermentation or physiological injury.

Applications include the storage and distribution of apples, grapes, berries, fresh-cut vegetables and salad products. These technologies reduce quality deterioration, help maintain firmness and freshness, and can reduce post-harvest losses. Their success depends on selecting suitable gas concentrations, packaging materials, storage temperatures and handling practices. MAP and CA do not replace hygiene, refrigeration or other necessary food safety measures.

30. Classify the major freezing methods based on mode of heat transfer and present suitable applications for each method.

Freezing preserves fruits and vegetables by lowering their temperature sufficiently to convert a substantial proportion of their water into ice. It slows microbial growth and many chemical and enzymatic reactions. Freezing methods can be classified according to the way heat is removed from the product.

Air-blast freezing uses rapidly circulating cold air to remove heat from food. It is suitable for fruits, vegetables and other products, especially when a flexible system is required. Fluidized-bed freezing is a form of air freezing in which small individual pieces, such as peas and diced vegetables, are suspended by a stream of cold air, allowing rapid and relatively uniform freezing.

Contact or plate freezing transfers heat through direct contact between packaged food and refrigerated metal plates. It is suitable for flat packages and products of uniform thickness. Immersion freezing involves placing suitable food or packages in a cold liquid medium, allowing rapid heat transfer; the medium must be appropriate for the product and packaging.

Cryogenic freezing uses very cold substances such as liquid nitrogen or carbon dioxide to freeze food rapidly. It is useful for high-value products, delicate fruits and individually quick-frozen items, although operating costs may be high.

The choice of method depends on product size, shape, packaging, required freezing rate, cost and final quality. Rapid freezing generally produces smaller ice crystals and can reduce tissue damage and drip loss during thawing. Correct packaging and stable frozen storage are necessary to maintain quality.

31. Explain the methods used to control post-harvest losses.

Post-harvest losses can be controlled through careful harvesting, suitable handling, proper storage, effective transportation and appropriate processing. Fruits and vegetables should be harvested at the correct maturity stage using methods that minimize bruising, cuts and crushing. Sorting and grading remove damaged, diseased and unsuitable produce and help maintain uniform quality.

Pre-cooling after harvest removes field heat and slows respiration and deterioration. Refrigerated storage at the appropriate temperature and relative humidity reduces moisture loss and delays ripening. However, chilling-sensitive commodities must be stored within their recommended temperature range. Suitable packaging protects produce from mechanical damage and reduces moisture loss. Refrigerated or well-managed transportation helps maintain the cold chain and prevents rapid quality deterioration.

Other methods include controlled atmosphere storage, modified atmosphere packaging and edible coatings, which can help slow respiration and moisture loss when properly applied. Sanitation, pest control and hygienic handling reduce microbial spoilage. Processing methods such as canning, freezing, drying, fermentation and preparation of jams and juices convert surplus or perishable produce into more stable products. Training handlers, improving storage infrastructure and maintaining accurate stock rotation further reduce losses. An integrated approach is most effective because post-harvest deterioration can occur at every stage between harvesting and consumption.

32. Discuss the factors affecting vinegar production, including temperature, acidity, oxygen supply and microbial activity.

Vinegar is produced mainly through the conversion of ethanol into acetic acid by acetic acid bacteria under aerobic conditions. The process generally involves two stages: alcoholic fermentation, in which yeast converts sugars into ethanol, followed by acetic acid fermentation, in which suitable bacteria convert ethanol into acetic acid.

Temperature influences microbial growth and the rate of fermentation. Each yeast and acetic acid bacterial culture has a suitable temperature range, and excessive heat may inhibit or damage the microorganisms. Therefore, temperature must be controlled according to the culture and production method.

Acidity increases as acetic acid accumulates. The acid concentration affects microbial activity and the progress of fermentation. Appropriate acidity and ethanol levels must be maintained to support the desired process without inhibiting the microorganisms prematurely.

Oxygen supply is essential during acetic acid fermentation because acetic acid bacteria require oxygen to oxidize ethanol into acetic acid. Adequate aeration supports efficient production, whereas insufficient oxygen can slow the process. However, excessive aeration may increase evaporation and create other operational difficulties.

Microbial activity depends on the type, quantity and condition of the yeast and acetic acid bacteria. Suitable cultures, hygienic equipment and the prevention of contamination help ensure consistent fermentation. The availability of ethanol, nutrients and suitable pH also influences microbial performance.

Other factors include the quality of the starting material, alcohol concentration, fermentation time, aeration method and sanitation. After fermentation reaches the desired acidity, the vinegar is clarified or filtered as required and packaged hygienically. Proper control of these factors helps produce vinegar with consistent acidity, flavour, quality and keeping properties.